Hao Li

Edinburgh, UK | +44 7942801680 | thealigrad@gmail.com

MSc Computer Science student with research and industry experience in multimodal learning and graph neural networks. Interested in AI/ML and NLP internship and graduate opportunities.

EDUCATION

University of Edinburgh **Edinburgh, UK**
MSc Computer Science *Sep 2026 - Present*
Selected courses: Accelerated Natural Language Processing, Machine Learning Systems, Machine Learning Practical.

North China Electric Power University **Beijing, China**
Bachelor of Intelligence Science and Technology | 86.45/100 *Sep 2022 - Jun 2026*
Selected courses: Machine Learning, Data Structures, Operating Systems, Database Systems, Python/R.

RESEARCH EXPERIENCE

Multimodal Sentiment Analysis with Graph Neural Networks
Project Lead *Jul 2024 - Dec 2024*
Summer Research Programme, National University of Singapore

- Designed a graph-based multimodal sentiment analysis framework using GNN encoders and cross-modal co-attention to model interactions within and across modalities.
- Implemented the full training and evaluation pipeline and conducted experiments.
- Led the project, with primary responsibility for model design, code implementation and manuscript writing.

AAAI Conference Paper
Co-author *Feb 2024 - Oct 2024*
Pioneer: Physics-informed Riemannian Graph ODE for Entropy-increasing Dynamics

- Reviewed graph ODEs, Riemannian geometry and physics-informed learning; derived and verified constrained Ricci flow and entropy-related formulations.
- Implemented and debugged core model components and the experimental pipeline for physics-informed graph learning.
- Contributed to manuscript writing, technical exposition and revisions of the theoretical and experimental sections.

Non-Contact Micro-Motion Measurement for Industrial Safety
Team Leader *Jun 2023 - Jun 2025*
National Undergraduate Innovation and Entrepreneurship Training Program

- Built a non-contact deformation measurement system based on 2D digital image correlation (DIC) for industrial safety monitoring.
- Designed a multi-stage analysis pipeline for full-field displacement and strain estimation, pixel-level deformation modelling and threshold-based early warnings.
- Produced registered software for high-risk industrial monitoring.

INTERNSHIP EXPERIENCE

Beijing Huayun Zhichuang Technology Co., Ltd.
Engineering Algorithm Intern Jul 2025 - Aug 2025

- Developed CAD functional modules using image processing and computer vision for engineering data analysis; extended DIC-based deformation analysis to CAD applications.
- Worked with R&D and product teams on technical integration, documentation and demonstration development.

360 Digital Security Group
Multimodal Algorithm Intern Jan 2025 - May 2025

- Designed and supported image-text multimodal algorithm solutions, focusing on cross-modal feature interaction and representation modelling.
- Conducted inference and experimental evaluation, analysed quantitative metrics and qualitative results, and developed demos with algorithm and product teams.

State Nuclear Electric Power Planning Design&Research Institute Co., Ltd.
AI Intern, Digital Management Center Jul 2024 - Nov 2024

- Contributed to an enterprise survey and design LLM using open-source frameworks to support automated engineering design review.
- Supported AI assistant development by integrating industry standards and engineering knowledge bases.

SKILLS

Programming: Python, C, C++, Java, HTML.
Technical experience: graph neural networks, multimodal learning, training and evaluation pipelines, image processing, DIC.
Languages: Mandarin Chinese (native); English (IELTS 7).